
CONVEXITY PROBLEM SHEET

1

(a) Starting from the definition of a convex function

$$f(ax + (1 - a)y) \leq af(x) + (1 - a)f(y) \quad (1)$$

Let $a = \epsilon/(x - y)$ and rearrange the inequality to give

$$(x - y) \left(\frac{f(y + \epsilon) - f(y)}{\epsilon} \right)$$

on the left-hand side. Taking the limit $\epsilon \rightarrow 0$ show that the function $f(x)$ lies above the tangent line $t(x) = f(y) + (x - y)f'(y)$ going through the point y .
[4 marks]

Rearranging the Equation 1

$$f(y + a(x - y)) \leq f(y) + a(f(x) - f(y)).$$

Or

$$\frac{f(y + a(x - y)) - f(y)}{a} \leq f(x) - f(y).$$

Letting $a = \epsilon/(x - y)$ then

$$(x - y) \frac{f(y + \epsilon) - f(y)}{\epsilon} \leq f(x) - f(y)$$

Taking the limit $\epsilon \rightarrow 0$ then using

$$\lim_{\epsilon \rightarrow 0} \frac{f(y + \epsilon) - f(y)}{\epsilon} = f'(y)$$

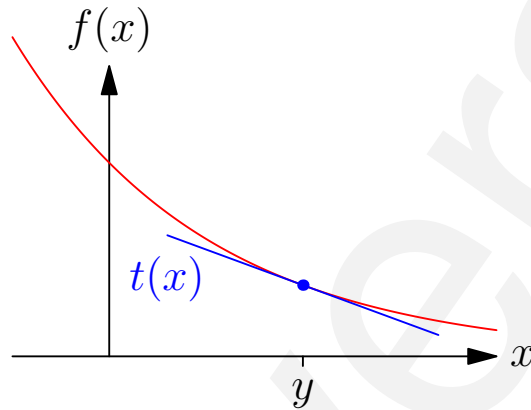
So that

$$(x - y)f'(y) \leq f(x) - f(y)$$

or

$$f(x) \geq f(y) + (x - y)f'(y) = t(x).$$

(b) Sketch the tangent line, $t(x)$, at the point y in the graph shown below. [1 mark]



(c) Starting from the inequality for a convex function

$$f(x) \geq f(y) + (x - y)f'(y) \quad (2)$$

consider the case $y = x + \epsilon$, then by Taylor expanding $f(x + \epsilon)$ and $f'(x + \epsilon)$ around x and keeping all terms up to order ϵ^2 show that for a convex function $f''(x) \geq 0$. [4 marks]

We use the expansions

$$f(x + \epsilon) = f(x) + \epsilon f'(x) + \frac{\epsilon^2}{2} f''(x) + O(\epsilon^3)$$

$$f'(x + \epsilon) = f'(x) + \epsilon f''(x) + O(\epsilon^2)$$

Using $y = x + \epsilon$ and substituting into the Equation (2)

$$f(x) \geq f(x + \epsilon) - \epsilon f'(x + \epsilon)$$

$$f(x) \geq f(x) + \epsilon f'(x) + \frac{\epsilon^2}{2} f''(x) + O(\epsilon^3) - \epsilon(f'(x) + \epsilon f''(x) + O(\epsilon^2))$$

Or subtraction $f(x)$ on both sides

$$0 \geq -\frac{\epsilon^2}{2} f''(x) + O(\epsilon^3)$$

Since this has to be true for all $\epsilon > 0$ this requires $f''(x) \geq 0$.

(d) Prove that $-\log(x)$ is convex-up for $x > 0$.

[1 mark]

$$\frac{d^2(-\log(x))}{dx^2} = \frac{1}{x^2} \geq 0$$

End of question 1

2

- (a) If $\|x\|$ is a proper norm use the triangular inequality ($\|x + y\| \leq \|x\| + \|y\|$), linearity of a norm ($\|ax\| = a\|x\|$) and the definition of convexity, to show that the norm is convex. [5 marks]

For any vectors, x , and y and any scale $a \in [0,1]$ then

$$\|ax + (1-a)y\| \leq \|ax\| + \|(1-a)y\| = a\|x\| + (1-a)\|y\|$$

where the first inequality follows from the triangular inequality and the second equality from the linearity of the norm. However,

$$\|ax + (1-a)y\| \leq a\|x\| + (1-a)\|y\|$$

is the defining equation of convexity.

- (b) Consider a classification problem where $\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})$ is the probability that a learning machine with parameters $\boldsymbol{\theta}$ predicts that input $\mathbf{x}$ belongs to class $c \in \mathcal{C}$. Assume the training is stochastic so the probability of obtaining parameters $\boldsymbol{\theta}$ is $\rho(\boldsymbol{\theta})$. Let $\hat{m}_c(\mathbf{x}) = \mathbb{E}_{\boldsymbol{\theta}}[\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})]$ be the output of the mean machine for class c . Assuming that for a data point $(\mathbf{x}, y)$, where y is a class label, we use a cross entropy loss

$$L(\mathbf{x}, y, \boldsymbol{\theta}) = - \sum_{c \in \mathcal{C}} \mathbb{I}[y = c] \log(\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})),$$

show that the expected loss over inputs and parameters can be written as the expected loss of the mean machine plus a second loss. Use Jensen's inequality ($\mathbb{E}[\log(X)] \leq \log(\mathbb{E}[X])$) to show the second term is positive. [5 marks]

$$\begin{aligned} \bar{L} &= \mathbb{E}_{(\mathbf{x}, y)} \left[\mathbb{E}_{\boldsymbol{\theta}} \left[- \sum_{c \in \mathcal{C}} \mathbb{I}[y = c] \log(\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})) \right] \right] \\ &= - \mathbb{E}_{(\mathbf{x}, y)} \left[\sum_{c \in \mathcal{C}} \mathbb{I}[y = c] \log(\hat{m}_c(\mathbf{x})) \right] - \mathbb{E}_{(\mathbf{x}, y)} \left[\mathbb{E}_{\boldsymbol{\theta}} \left[\sum_{c \in \mathcal{C}} \mathbb{I}[y = c] \log\left(\frac{\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})}{\hat{m}_c(\mathbf{x})}\right) \right] \right] \end{aligned}$$

The first terms acts like a bias. The second (variance-like) term is

$$- \mathbb{E}_{(\mathbf{x}, y)} \left[\sum_{c \in \mathcal{C}} \mathbb{I}[y = c] \mathbb{E}_{\boldsymbol{\theta}} \left[\log(\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})) \right] \right] + \mathbb{E}_{(\mathbf{x}, y)} \left[\sum_{c \in \mathcal{C}} \mathbb{I}[y = c] \log(\hat{m}_c(\mathbf{x})) \right]$$

But using Jensen's inequality

$\mathbb{E}_{\boldsymbol{\theta}} \left[\log(\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})) \right] \leq \log(\mathbb{E}_{\boldsymbol{\theta}}[\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})]) = \log(\hat{m}_c(\mathbf{x}))$. Thus this second term is positive.

End of question 2

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